

TCS NQT PYQ QUESTIONS

Company: TCS NQT **Round:**

Online

Year: 2025

Difficulty: Medium-Hard

Topic: Time and Work

Source: <https://bharatnqt.com/tcs-nqt-previous-year-questions/>

Problem Description:

A can complete a work in 20 days, B in 30 days, and C in 60 days. A and B work together for 5 days. Then B leaves and C joins A. After another 5 days, A leaves. How many more days will C take to finish the remaining work?

Options:

A) 16 days

B) 18 days

C) $144/7$ days

D) 12 days

Correct Answer:

C) $144/7$ days

Approach to Solve This Problem:

STEP 1:

A's 1-day work:

$$1/18$$

B's 1-day work:

$$1/24$$

Together:

$$1/18 + 1/24 = 7/72$$

Work completed in 6 days:

$$6 \times 7/72 = 7/12$$

STEP 2:

Remaining work: $1 - 7/12 = 5/12$

A + C complete $\frac{5}{12}$ work in 4 days.

Therefore, their 1-day work:

$$\frac{5}{12} \div 4 = \frac{5}{48}$$

C's 1-day work:

$$\frac{5}{48} - \frac{1}{18} =$$

$$(\frac{15-8}{144}) = \frac{7}{144}$$

Therefore, C alone takes:

$$144/7 \text{ days}$$